

AP Chem Unit 6: Electrochemistry



U6D1 January 6-7 (T/W)

HAPPY NEW YEAR! With your new tablemates:

Can you translate these Gen ALPHA New Year's Resolutions and apply them?

1. Be a sweat silently. A nonchalant W is the best W.
2. Less yapping, more executing on those big brain thoughts.
3. If it's giving crash out, touch grass and reset.
4. Don't let the ops live rent free in your head.
5. Lock in, don't crash out, put the fries in the bag.
6. Low key no chopped behavior, only top tier, elite, tough actions.
7. Discipline builds steez.
8. Cook consistently, not occasionally.
9. Nonchalant effort beats performative energy every time.

U6D1 January 6-7 (T/W) -

Learning Target: What is a redox reaction?

Agenda:

- ~New seats! New Stickers! New Unit: Electrochemistry
- ~Electrochem day 1! How do we assign oxidation numbers?
- ~Oxidation and Reduction POGIL (Lab Grade- let's start off with a 100!)

Homework:

- ~Finish the Redox POGIL
- ~Watch the EdPuzzle by 1/12-13
- ~Start Textbook Problem Set #1
- ~Quiz on 1/16-20
- ~Test on 1/23-26

U6D2 January 8-9(Th/F) -

Learning Target: How are redox reactions balanced? What is a redox titration?

Agenda:

- ~WARM UP! Turn to pg 3 of your packet and try the practice at the bottom of the page
- ~Review POGIL
- ~Redox RXNs- determining and balancing half reactions
- ~Redox Titration- it's just fancy electric stoich....
- ~[Redox Titration Simulation](#)

Homework:

- ~Finish Redox Titration by 1/14-15
- ~Textbook Problem Set #1
- ~Quiz on 1/16-20
- ~Test on 1/23-26

U6D3 January 12-13 (M/T) -

Learning Target: What is a galvanic cell? How do we calculate cell potential?

Agenda:

- ~Warm Up: As a table, solve the redox titration problem on p 8 of your packet
- ~Galvanic cells: how do they work?
- ~Calculating Cell Potential

Homework:

- ~Finish Redox Titration Sim if needed
- ~Textbook Problem Set #1 and 2?
- ~Quiz on 1/16-20
- ~Test on 1/23-26

U6D4 January 14-15 (W/Th) -

Learning Target: How do we calculate cell potential?

Agenda:

~Warm Up- Grab a big whiteboard for the table:

~Calculating Cell Potential- Notes and Practice

~Spontaneity and electrochemical cells

Homework:

~Quiz next time

~Prob set #1 due at 11:55 pm tomorrow! Prob set #2 is posted

~Test on 1/23-26

~Warm Up- Grab a big whiteboard for the table:

For the half reactions: $\text{Al} \rightarrow \text{Al}^{+3} + 3\text{e}^-$ and $\text{Fe}^{+2} + 2\text{e}^- \rightarrow \text{Fe}$

- Determine the complete redox reaction.
- Draw a cell that represents the reaction. Label the anode and the cathode and the electron flow. Include a KNO_3 salt bridge and a voltmeter.
- Which electrode gains mass? Loses mass? In the salt bridge, which way do the cations flow? The anions?
- The cell voltage, E_{cell} , is calculated: $E_{\text{cell}} = E_{\text{cat}} - E_{\text{an}}$. Use the values on the back of your packet to determine the cell voltage.

U6D5 January 16-20 (F-T) -

Learning Target: How are cells at nonstandard conditions different?

Agenda:

~Quiz!

~More on Spontaneity and Ecell

~Cells at nonstandard conditions

Homework:

~Textbook Problem Set #2

~Wear your CLOSED TOED SHOES next time

~Test on 1/23-26

U6D6 January 21-22 (W/Th) -

Learning Target: Can we measure cell voltage?

Agenda:

- ~Finish the packet! Electrolytic cells

- ~Lab!

- ~Practice FRQ

Homework:

- ~Textbook Problems due next time

- ~Test next time

- ~U7 Acids/Base Notes for p 1-6 (first 2 recorded lectures) due w open note quiz

U6D7 AND U7D1 January 23-26 (F/M) -

Learning Target: What have you learned about Electrochem?
What are Arrhenius and Bronsted-Lowry acids and bases? How do we calculate pH?

Agenda:

- ~Get ready for the U6 Test!
- ~New Unit! U7: Acids and Bases- pg 1-6- listen to the first 2 lectures (Intro to A/B and pH & pOH of strong A/B)
- ~Open note quiz over A/B p1-6

Homework:

- ~Unit 7 Textbook Prob Set 1 going up shortly
- ~Unit Exam 2/20-23

